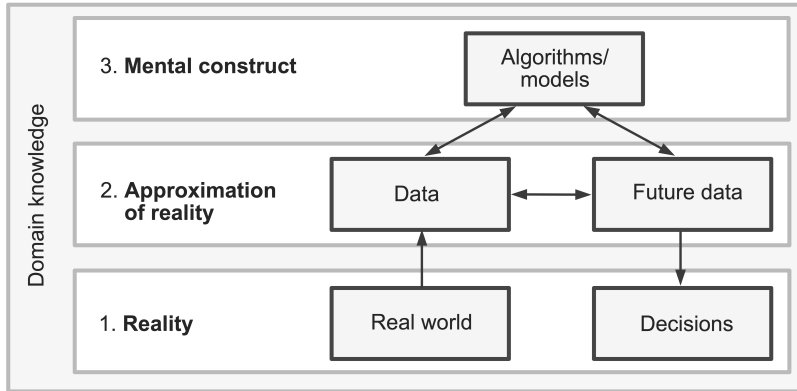


Model diagnostics

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Question: Is our model useful?

Model assessment

We observe data $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$ and fit the model

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

How do we assess whether the model is useful?

- Does the model do a good job predicting the response?
cross-validation, data splitting, etc.
- Do the model assumptions to (at least approximately) match the observed data?
diagnostic plots

Model assumptions

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

What assumptions do we make when fitting this logistic regression model?

- Distribution : $Y_i | X_i$ is Bernoulli ✓
- Shape : log-odds are linear function of x_i ?
- Independent observations (x_i, y_i)
⇒ likelihood is product over n observations
- Randomness : data come from random process
(random sample, randomized experiment, etc.)
- Lack of outliers (all observations come from same process)
- Lack of multicollinearity

Shape assumption

$$Y_i | \mathbf{x}_i \sim \text{Bernoulli}(p_i)$$
$$\log \left(\frac{p_i}{1 - p_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}$$

Shape assumption: The log-odds are a linear function of the explanatory variables (i.e., the systematic component is correctly specified).

Question: How do we assess the shape assumption in a *linear* regression setting?

residual plots,

Residual plot?

Raw residuals: $Y_i - \hat{p}_i$

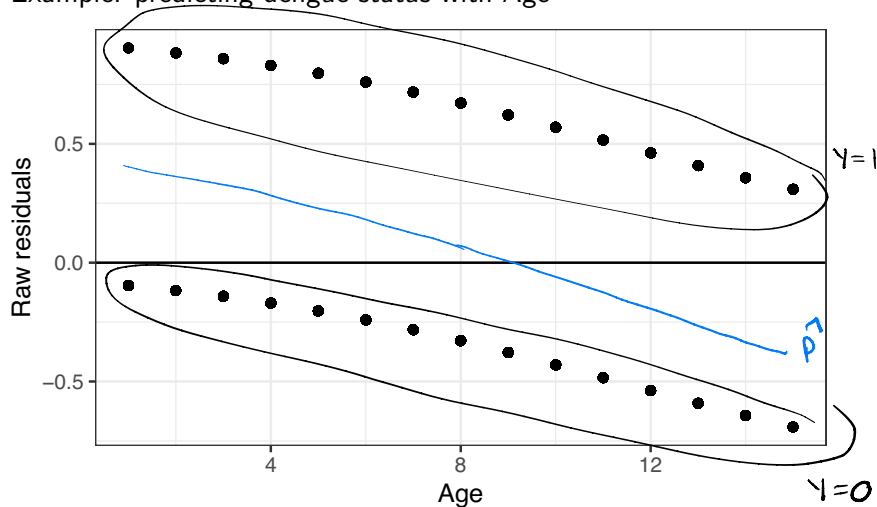
Question: what will the plot look like if we plot $Y - \hat{p}$ vs. X ?

Residual plot?

Raw residuals: $Y_i - \hat{p}_i$

Question: what will the plot look like if we plot $Y - \hat{p}$ vs. X ?

Example: predicting dengue status with Age



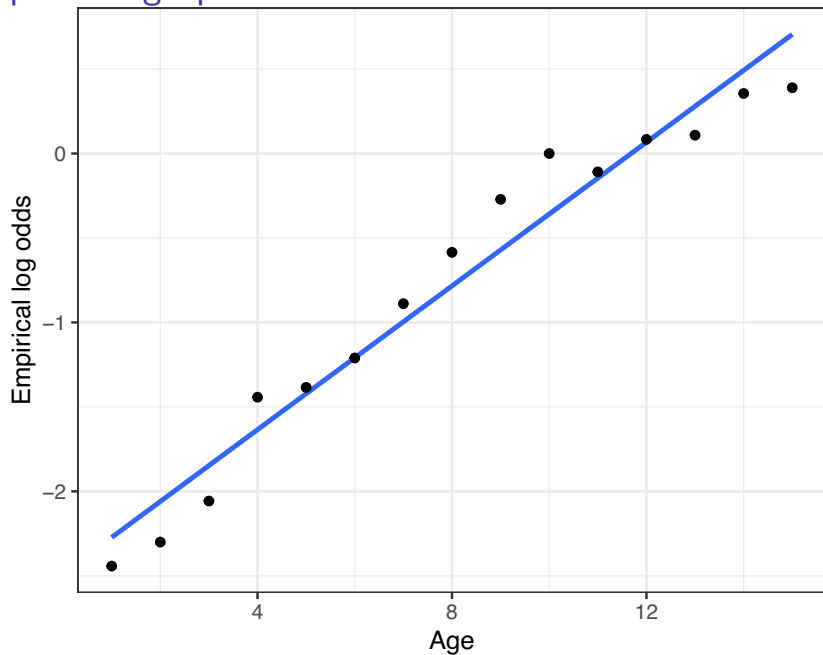
Alternative: empirical logits

For each Age, calculate the *empirical* log odds of dengue (aka *empirical logits*):

```
## # A tibble: 15 x 5
```

##	Age	NoDengue	YesDengue	p	log_odds
##	<int>	<int>	<int>	<dbl>	<dbl>
##	1	1	184	16 0.08	-2.44
##	2	2	419	42 0.0911	-2.30
##	3	3	532	68 0.113	-2.06
##	4	4	457	108 0.191	-1.44
##	5	5	487	122 0.200	-1.38
##	6	6	369	110 0.230	-1.21
##	7	7	360	148 0.291	-0.889
##	8	8	289	161 0.358	-0.585
##	9	9	244	186 0.433	-0.271
##	10	10	185	185 0.5	0
##	11	11	165	148 0.473	-0.109
##	12	12	126	137 0.521	0.0837
##	13	13	96	107 0.527	0.108

Empirical logit plot



Empirical logits with a continuous variable

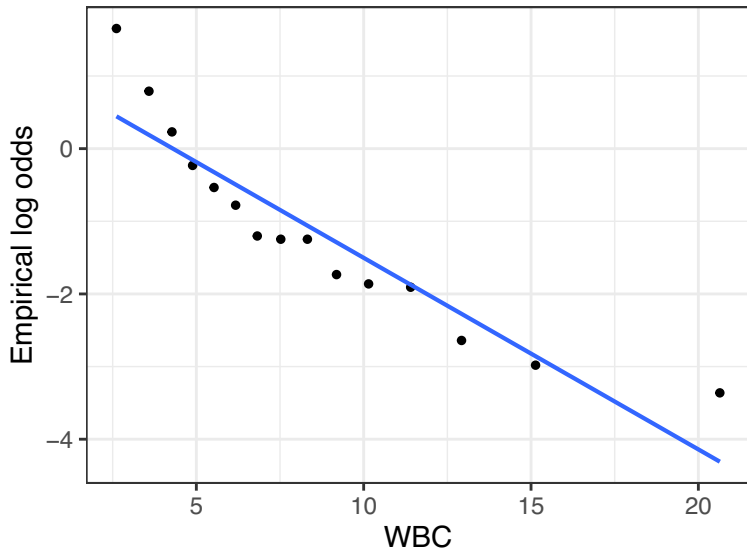
Trying with WBC:

```
## # A tibble: 6 x 5
##       WBC NoDengue YesDengue     p log_odds
##   <dbl>   <int>    <int> <dbl>   <dbl>
## 1  0.9         0        1     1     Inf
## 2  1.15        0        2     1     Inf
## 3  1.23        0        1     1     Inf
## 4  1.25        0        1     1     Inf
## 5  1.54        0        3     1     Inf
## 6  1.58        0        1     1     Inf
```

What has gone wrong? What should we do differently?

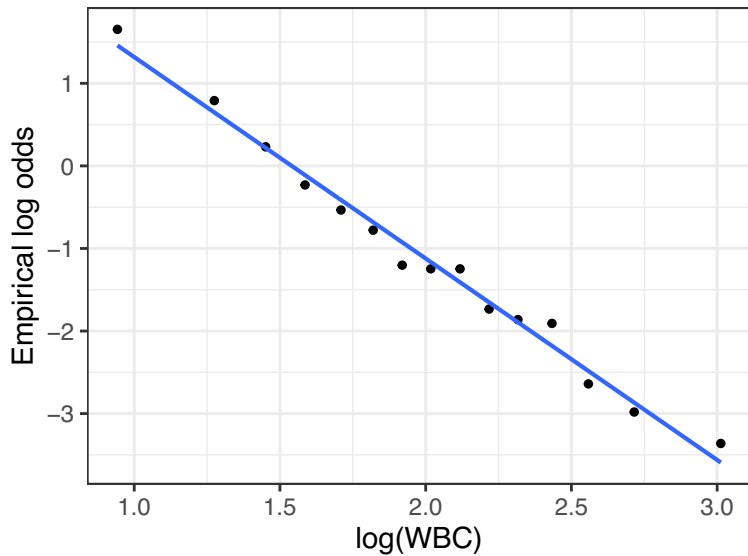
Bin (group) observations by WBC, calculate
empirical log odds for each bin

Binned empirical logit plots

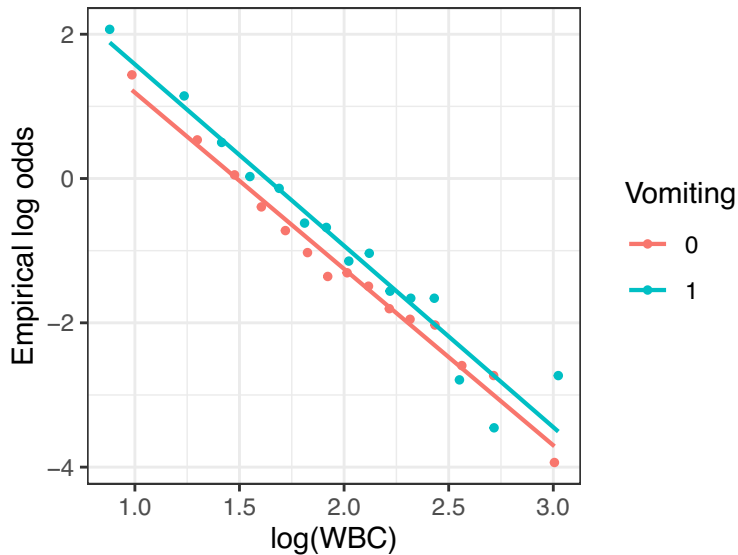


Does the shape assumption seem reasonable?

Binned empirical logit plots



Binned empirical logit plots



Limitations of empirical logit plots

- ▶ Does not assess a fitted model (performed *before* fitting the model)
- ▶ One point per *bin*, not one point per *observation*
- ▶ Can only plot one or two explanatory variables at a time

Randomized quantile residuals

Define randomized quantile residuals $r_{Q,i}$:

Motivation: Suppose $\hat{p}_i = 0.8$. Want:

if $\hat{p}_i \approx p_i$ (good estimate), then $E[r_{Q,i} | x_i] \approx 0$

if $\hat{p}_i > p_i$ (overestimate), then $E[r_{Q,i} | x_i] < 0$

if $\hat{p}_i < p_i$ (underestimate), then $E[r_{Q,i} | x_i] > 0$

And want $r_{Q,i} \approx \text{Normal}$ if $\hat{p}_i \approx p_i$

Idea:

$\hat{p}_i = 0.8$

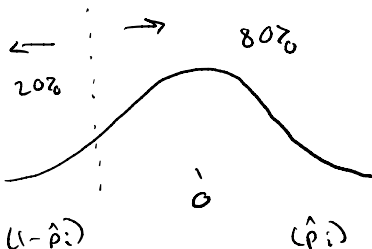
Divide $N(0,1)$ into 2 regions

if $\gamma_i = 1$, sample $r_{Q,i}$ from upper part

if $\gamma_i = 0$, sample $r_{Q,i}$ from lower part

if $\hat{p}_i \approx p_i$, then $P(\gamma_i = 1) \approx \hat{p}_i$

so on average, I'm sampling from a $N(0,1)$!

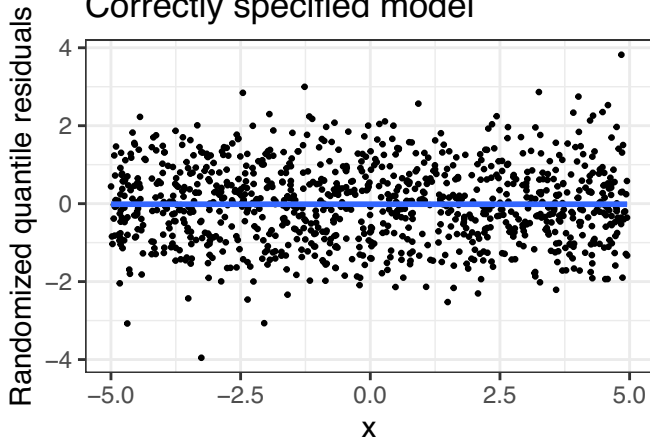


Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

Correctly specified model



Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i^2$

Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

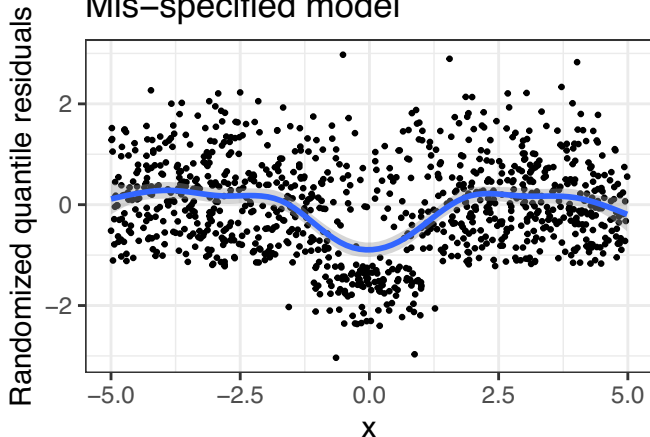
Question: What do you think the randomized quantile residual plot will look like?

Randomized quantile residuals

Simulated data: $\log\left(\frac{p_i}{1-p_i}\right) = -1 + 2X_i^2$

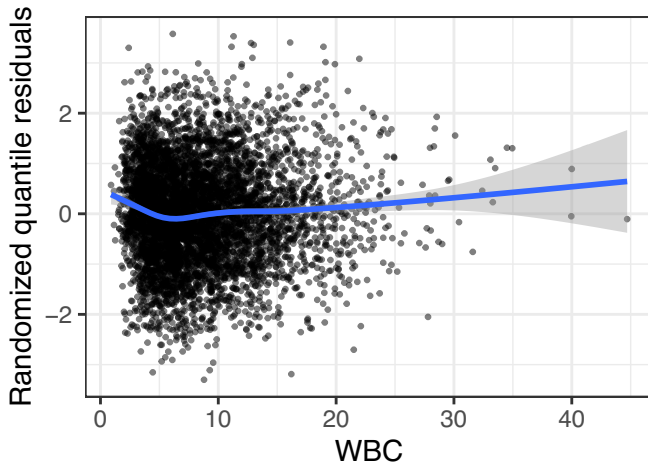
Model fit: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_i$

Mis-specified model



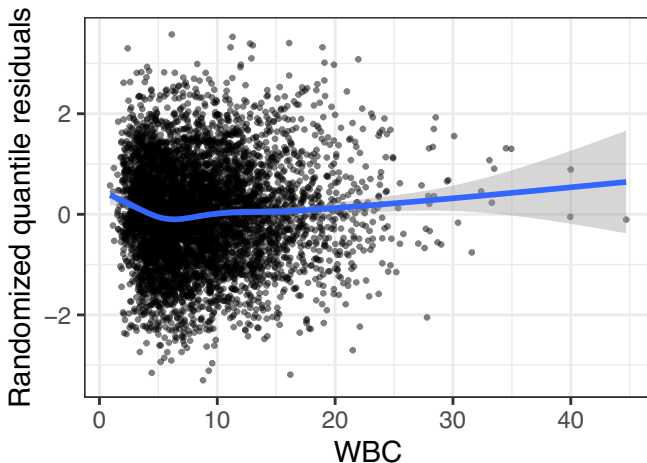
Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



Example: dengue data

Model: $\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 \text{WBC}_i$



Question: How can we decide whether a plot is consistent with a correctly-specified model?

Parametric bootstrap

The *parametric bootstrap* generates data from the *fitted* model:

· Given data $(x_1, y_1), \dots, (x_n, y_n)$, fit model
 $\rightarrow \hat{p}_1, \dots, \hat{p}_n$

· simulate new responses: $y_1^*, \dots, y_n^*$
 $y_i^* \overset{\text{indep.}}{\sim} \text{Bernoulli}(\hat{p}_i)$

(Now we know that the model is correctly specified for y^*)

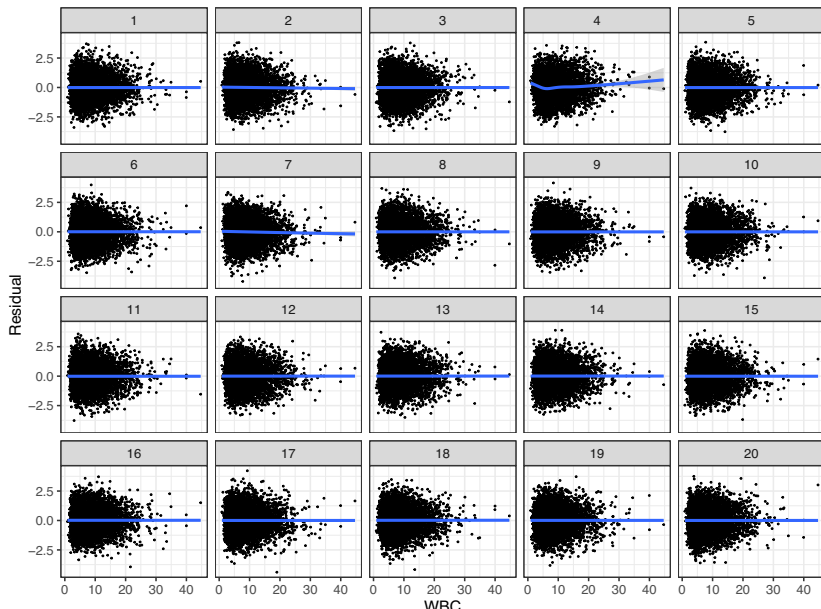
· Re-fit model / diagnostic / etc. on
 $(x_1, y_1^*), \dots, (x_n, y_n^*)$

Class activity

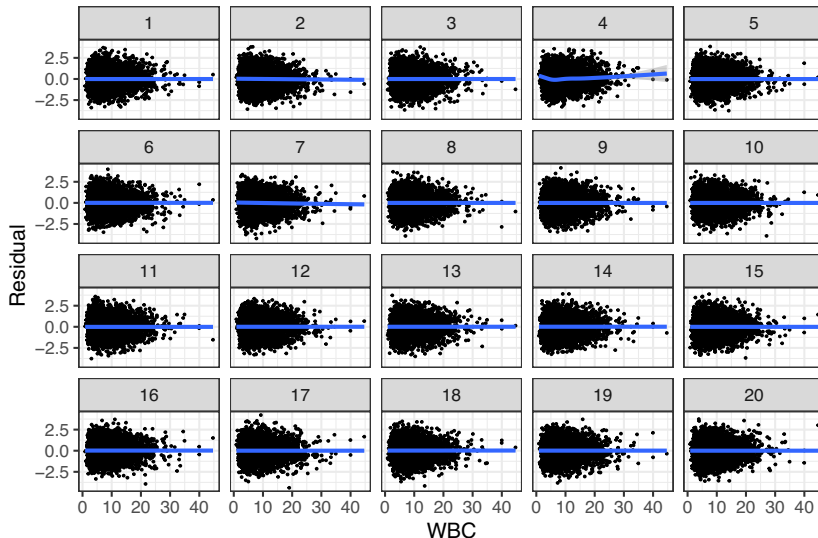
Work on the class activity (available on the course website).

Visual inference

```
## decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```



Visual inference



```
decrypt("QUg2 qFyF Rx 8tLRyRtx Ko")
```

```
## [1] "True data in position 4"
```